Mehul Tomer

B.Tech Computer Science Graduate | Full-Stack Development & AI/ML Enthusiast
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ABOUT ME

A Computer Science (B.Tech) graduate with hands-on experience building AI-integrated projects and a strong interest in modern AI automation tools. Implemented machine learning models, computer vision pipelines, and LLM API integrations across academic and personal projects, gaining practical exposure to tools like LangChain, OpenAI API, and SHAP. Familiar with trending AI workflows including agentic pipelines and RAG — explored these during project work and self-learning. Eager to bring this foundational AI exposure into a professional setting and grow further within an innovative team.

EDUCATION

Bachelor of Technology in Computer Science and Engineering
SRM Institute of Science and Technology | 09/2021 – 05/2025 | Chennai, India | CGPA: 9.36

Higher Secondary Certificate (HSC)
SR DAV Public School | 2019–2020 | Delhi, India | Grade: 9.20

WORK EXPERIENCE

SDE Intern – eGIANTS | Chennai, India | Jul 2025 – Oct 2025

- Implemented robust, scalable RESTful APIs using Node.js to manage back-end data and ensure seamless communication between client-side applications and databases.
- Developed responsive user interfaces for a major publishing and learning multimedia client platform, enhancing multi-device and multi-platform usage.

Project Trainee – RINL Vizag Steel Plant | Visakhapatnam, India | Aug 2024 – Sep 2024

- Executed a project on Database Management Systems focusing on Oracle 11g and SQL.
- Collaborated with senior engineers on real-world industrial database challenges and documented technical processes and user manuals.

PROJECTS

Eco-Friendly Vehicle Detection & Carbon Emission Analysis [Jun 2025]

- Built a real-time vehicle detection and classification pipeline using YOLOv5s and OpenCV, trained on the COCO2017 dataset to categorise vehicles as eco-friendly or non-eco-friendly.
- Implemented a LangChain-based automation workflow connecting the detection output to an OpenAI API call, generating a natural language carbon emission summary report per detected vehicle — hands-on exploration of agentic AI pipelines.
- Used CometML to log and visualise training metrics in real time, gaining familiarity with experiment tracking tools used in ML workflows.

Tools: *Python, YOLOv5s, OpenCV, LangChain, OpenAI API, CometML, NumPy, Pandas, Jupyter Notebook*

Link: github.com/mehu11/Eco_friendly_Vehicle_Detection_using_yolo_v5

Chaos Theory Explorer – Logistic Map Visualiser with AI Q&A [Apr 2025]

- Simulated and visualised the logistic map to study nonlinear dynamics using Python, implementing bifurcation diagrams and time-series analysis to demonstrate transitions between order and chaos.
- Built an interactive Streamlit web app for real-time parameter exploration; integrated an OpenAI API-powered chatbot that answers user questions about the graphs, applying a basic RAG-style approach to ground responses in context.
- Implemented with scikit-learn for supporting data analysis; the project was a practical introduction to combining data visualisation apps with LLM-based Q&A.

Tools: *Python, NumPy, Matplotlib, Streamlit, scikit-learn, OpenAI API, Git*

Link: Project Drive Link

T20 World Cup Score Predictor [Dec 2024]

- Developed a regression model using XGBoost and scikit-learn to predict final T20 match scores from in-game situational data; performed feature engineering and data preprocessing with Pandas and NumPy.
- Integrated SHAP (SHapley Additive exPlanations) to make the model interpretable — visualising which features like run rate and wickets most influenced each prediction.
- Built a Streamlit front-end for real-time predictions; explored using GPT API to convert SHAP output into plain-English match insights, experimenting with LLM-as-explainability-layer pattern.

Tools: *Python, XGBoost, SHAP, OpenAI API, Streamlit, NumPy, Pandas, scikit-learn, Jupyter Notebook, Git*

Link: github.com/mehu11/T20_-world_cup_winner_predictor

SKILLS

AI / ML	YOLOv5s · scikit-learn · SHAP · NumPy · Pandas · Matplotlib
AI Tools & Automation	LangChain · OpenAI API · Anthropic Claude API · RAG (basic) · Agentic Pipelines (basic)
Backend & APIs	Node.js · Spring Boot · RESTful APIs · FastAPI · Docker
Frontend	React.js · HTML · CSS
Languages	Python · JavaScript · C++ · SQL
Tools & Platforms	Git · GitHub
Soft Skills	Teamwork · Analytical Thinking · Attention to Detail · Strategic Thinking

PUBLICATIONS

Eco-friendly Vehicle Detection and Carbon Emission Analysis using YOLOv5s

Mehul Tomer, et al. | ICNEXT'25 – International Conference On Next-Gen Engineering And Smart Technologies | 2025 (Manuscript submitted for publication)

Proposed a real-time deep learning system combining YOLOv5s and image processing to classify vehicles by energy type and estimate carbon footprint, enabling data-driven sustainable urban transportation planning.

Link: View Manuscript

LEADERSHIP & VOLUNTEERING

Technical Lead – Techwiz Club, SRM University | Sep 2023 – Sep 2024

- Led workshops and seminars; mentored junior members in programming, marketing, and management.
- Technical Coordinator for PAIR-A-THON inter-college coding event attracting 100+ participants; designed problem statements and managed a volunteer team.

Volunteer Activities

- Tree Plantation Drive (Sep–Oct 2024, New Delhi) – promoted environmental sustainability and peer engagement.
- Sawastik Sewa Foundation NGO (Dec 2023–Jan 2024) – hygiene and sanitation awareness campaign in underserved communities.

LANGUAGES

Hindi (Mother Tongue) | English (Proficient – C1: Listening, Reading, Writing, Speaking)